IILM UNIVERSITY

IILM University, Greater Noida

SCHOOL OF COMPUTER SCIENCE & ENGINEERING

PROJECT APPROVAL FORM AND ABSTRACT
Even 2026-2027

B.Tech Project Details

Project ID: BTP2CSE065

Project Title	Machine Learning Based Network Load Prediction for Edge Server		
Project Type	☐ Community-Based Design Problem (Interdisciplinary) ☐ App Development/Utility ☐ IOT/Hardware-based ☑ AI/ML/Data Science ☐ Healthcare Projects ☐ Cyber Security and networking	Project Outcome	☑ Project and Research Paper ☐ Project and Patent ☐ Project and Book Chapter
		SDG Goal Number	9
Publication Target	☐ SCOPUS Journal ☑ SCOPUS Conference ☐ SCI Journal ☐ SCOPUS Book Chapter	Guide Name: Dr. Prakhar Consul	

Student Details:

S. No	Name	Roll No.	Admission No.	B.Tech/Branch	Sem
1	Vijendra kumar	2410030862		B.Tech (CSE)	4th
2	Rajyavardhan	2410030715		B.Tech (CSE)	4th
3	Shourya Singh	2410030816		B.Tech (CSE)	4th
4	Vicky Raj	2410030749		B.Tech (CSE)	4th
5	Mayank Bhardwaj	2410030870		B.Tech (CSE)	4th

Sustainable Development Goals (SDGs): Targets and Indicators

Goal 1: Eliminate poverty in all its manifestations across the world
Goal 2: Eradicate hunger, enhance food security and nutrition, and support sustainable agricultural practices
Goal 3: Promote good health and well-being for people of all ages
Goal 4: Provide equitable and inclusive quality education and encourage lifelong learning opportunities

Goal 5: Attain gender equality and empower women and girls in all spheres
Goal 6: Guarantee access to clean water and ensure sustainable sanitation management
Goal 7: Ensure the availability of affordable, dependable, sustainable, and modern energy sources
Goal 8: Foster inclusive and sustainable economic development, productive employment, and decent work opportunities
Goal 9: Develop resilient infrastructure, encourage sustainable industrial growth, and promote innovation
Goal 10: Minimize inequalities within nations as well as across countries
Goal 11: Create cities and communities that are inclusive, safe, resilient, and sustainable
Goal 12: Promote responsible consumption and sustainable production practices
Goal 13: Take immediate and effective measures to address climate change and its consequences
Goal 14: Protect and sustainably manage marine and ocean resources
Goal 15: Conserve and restore land ecosystems, manage forests responsibly, and prevent land degradation and biodiversity loss
Goal 16: Encourage peaceful and inclusive societies, ensure justice for all, and strengthen transparent and accountable institutions
Goal 17: Enhance global cooperation and strengthen partnerships to achieve sustainable development objectives

Abstract This project is about predicting the network load on edge servers using machine learning. Nowdays, because of IOT devices and real-time applications, edge servers receive a large and changing amount of network traffic. Traditional load management methods works only after a problem occurs, which can cause server overload and poor performance.

In this project, past network data such as CPU usage, memory usage, and network traffic are used to train machine learning models. By predicting the future load in advance, the system can manage resources better and distribute the load efficiently. This helps in improving server performance and reducing delay in the system.

Problem Statement

Edge servers faces changing network traffic because user demand varies over time, which can cause server overload and delays. Existing load management methods do not predict future traffic effectively.

This project aims to use machine learning to predict network load on edge servers using past data, helping in better load distribution and efficient use of resources.

Signature of Students:

Rajyavardhan
Vicky Raj
Vijendra kumar
Shourya Singh.
Mayank

Signature of Guide:

10/02/26